

SANJEEVANI — Production Cisco Network Architecture

Maps the POC (Docker containers simulating links) onto real Cisco hardware. Every element below corresponds to something already working in the codebase — the hardware doesn't replace the software story, it embodies it.

1. Product Selection — Scored Against the Problem Statement

The problem statement needs exactly ten capabilities. Each selected product earns its place against one of them; everything else from the catalog is rejected (Section 6) to keep the design credible.

#	Requirement	Selected Cisco Product	Where it already exists in code
1	Citizens connect with zero surviving infrastructure	Catalyst IW9167E heavy-duty AP (IP67) — open captive-portal Wi-Fi bubble	Pod React UI, <code>ciscoSimulation.localWifi</code>
2	Pod runs its own brain with no server rack	Catalyst IR1800 rugged router + Cisco IOx running the pod-agent Node.js container + local queue on flash	<code>pod-agent/server.js</code> , <code>ciscoSimulation.podEdge</code>
3	Multi-path WAN failover (the ladder)	IR1800 WAN ethernet → Starlink LEO , 5G/LTE pluggable module , optional P-LTE-450 private LTE	<code>connectivityManager.calculateMode()</code>
4	Pod-to-pod mesh when both WANs die	Catalyst IW9165E with URWB (directional, km-range, fiber-like)	<code>gossipRouter.js</code> , <code>/api/mesh/inbox</code>
5	A lifeline when even the mesh dies	IXM-LPWA LoRaWAN gateway — 10–15 km sub-GHz thin pipe	<i>New tier</i> — extends island mode (see 4.5)
6	Power + connect everything in the pod	Catalyst IE3400 industrial PoE switch (fanless, one battery powers all antennas)	implicit in pod topology
7	Sensor-driven early warning	Meraki MT10/MT12/MT14 sensors + MT30 button	<code>hazardPackService.js</code> , <code>sensor-simulator</code> , <code>/api/sensors/button</code>
8	Tamper-proof alerting + responder notification	App-layer Ed25519 (ours) over SD-WAN IPsec + Webex bot	<code>verifyAlert()</code> , <code>webexNotifier.js</code>
9	EOC visibility, control, pre-failure warning	Catalyst SD-WAN (vManage) + ThousandEyes IOx agents + IoT Operations Dashboard / Meraki Dashboard	<code>command-center</code> , <code>link-node /health</code> degraded signal, <code>simulation-controller</code> topology
10	Field humans must still talk in island mode	IP Phone 8800 + SRST on the IR1800 (router becomes the local exchange)	<i>New</i> — complements coordinator guild halls

Strong optional additions (include if asked "what else"): - **Meraki MV72X** rugged camera — edge AI, on-camera storage, sends a 50 KB snapshot instead of a stream: the physical embodiment of our store-and-forward law. - **Catalyst Micro switches** — backpack pods extending Wi-Fi into collapsed structures. - **Cisco Cyber Vision** on the IE3400 — OT security for the sensor fleet. - **Secure Equipment Access (SEA)** — zero-trust remote config of pod routers without VPN.

EOC (the Keep): 2× **UCS C220** rack servers (cloud-api, MongoDB, Ollama, command-center — our whole compose file), behind a **Secure Firewall 1150**, on a **Catalyst 9300** access switch. Cloud-hosted SD-WAN controllers and Webex Control Hub. Deliberately small — the EOC is a tent, not a data center.

2. The Architecture — Five Planes



The five planes:

- 1. Access plane** — IW9167E captive portal (open SSID by design: a victim must connect with zero credentials), MT sensors, MT30 buttons, MV72X snapshots.

- 2. **Compute plane** — IR1800/IOx runs the pod-agent exactly as it exists today; `queue.json` lives on router flash. The pod *is* the router.
- 3. **Transport plane** — the four-tier ladder (Section 3).
- 4. **Control plane** — SD-WAN policy = our `calculateMode()` in silicon; RESTCONF zero-touch provisioning (already proven live by `integrations/catalyst_restconf.py` against a real Catalyst 8000v); IoT Ops Dashboard + SEA for fleet management.
- 5. **Observability & security plane** — ThousandEyes agents *inside* the IR1800 via IOx; SD-WAN IPsec below, our Ed25519 alert signing above (defense in depth: even a compromised transport can't forge an evacuation order); Umbrella DNS filtering on the captive portal; Cyber Vision watching the OT sensors.

3. The Transport Ladder — Code ↔ Hardware

Tier	POC today	Production hardware	Trigger to fall to next tier
1 Satellite	<code>link-node :9100</code> (80 ms, loss dice)	Starlink/LEO on IR1800 WAN1	ThousandEyes path test: loss ≥ 25% or dish offline → SD-WAN demotes color
2 Cellular	<code>link-node :9201/:9202</code> (30 ms)	5G module on IR1800; P-LTE-450 where commercial towers are dead (450 MHz = massive range + wall penetration, private emergency spectrum)	tower unreachable / SIM registration lost
3 Mesh	<code>gossipRouter</code> + <code>/api/mesh/inbox</code>	IW9165E URWB directional bridges, pod→pod up to ~5 km; URWB's make-before-break ≈ our BGP-path-vector gossip in hardware	no URWB neighbor with an upstream path
4 LoRa thin pipe	(new — not in POC)	IXM-LPWA : ~242-byte payloads, 10–15 km through rubble	n/a — always on, in parallel
5 Island	<code>queue.json</code> + syncWorker retry	IOx flash storage + MV72X on-camera recording + SRST local voice	—

Tier 4 is the one genuinely new capability the hardware adds to the software story: even a fully islanded pod whispers a heartbeat — `POD-3: Q-45, CRIT-2` (pod id, queue depth, critical count) — every 30 s over LoRa. The EOC dashboard shows *"island, but alive, 45 queued, 2 critical"* instead of a grey square. A compressed critical SOS (id + category + severity + GPS ≈ 200 bytes) can even trickle out one per uplink. This upgrades island mode from "dark but safe" to "dark but heard."

4. Communication Flows — All the Possibilities

A. Normal operations. Victim's phone → IW9167E captive portal → IE3400 → IR1800, where the IOx pod-agent triages (11 languages, offline) and queues to flash → SD-WAN IPsec tunnel over Starlink → Secure Firewall → cloud-api on UCS → AI triage (Ollama on UCS) → coordinator delivery → Webex push to responder phones. *Same nine steps as the demo; only the containers became metal.*

B. Degradation (predictive failover). Rain fade hits the dish. The ThousandEyes agent inside the IR1800 sees loss climbing on the satellite path *before* users feel it and reports path quality; SD-WAN application-aware routing demotes the satellite color and pins SOS-class traffic to cellular. Telemetry class may stay on the degraded satellite (it can afford loss). This is `DEGRADED_LOSS = 0.25` and the "degraded-still-beats-island" tail of `calculateMode()`, executed in silicon. **QoS note:** SOS traffic is DSCP-marked EF at the IR1800; the IE3400 and SD-WAN honor it — the production form of our priority lanes and batch sync.

C. Total WAN loss → mesh. Flood takes the dish and the towers. The IW9165E URWB radios already hold formed links to neighbor pods; the IR1800's route to the EOC now points across the mesh to POD-05, whose own SD-WAN tunnel over its surviving tower carries both pods' traffic. Our `relayTrail` bookkeeping remains at the application layer, giving the EOC the full provenance of every relayed SOS.

D. True island + thin pipe. POD-06 (no mesh neighbor, per our topology). Everything queues on flash — Law 1, unchanged. The IXM-LPWA whispers the heartbeat to the EOC's LoRaWAN network server 12 km away. The MV72X records locally and queues 50 KB AI-detected snapshots ("person detected, 14:32") behind the SOS class. The medic picks up the 8800 phone and dials the shelter coordinator — **SRST** on the IR1800 switches the call locally over the pod's own network. The pod is cut off and still functioning as a tiny civilization.

E. Restoration. Tower comes back. ThousandEyes confirms the path, SD-WAN restores the color, the pod-agent's health-change listener fires, and the syncWorker drains the queue — batched (`forward-batch`) exactly as trial [7] of `integration_test.py` proves today. Delayed, never lost.

F. Signed alert broadcast (downstream). EOC issues an evacuation order → cloud-api signs (Ed25519, seq, scope) → fan-out down every SD-WAN tunnel, across URWB to meshed pods, and — as a 180-byte compressed form — over LoRa to islanded pods. Every pod verifies with its cached public key **regardless of which pipe delivered it**. A forged alert injected on any radio layer still dies at the pod with a 401 + SECURITY event. Transport security (IPsec) and message security (Ed25519) are independent layers — that's the pitch line.

G. Management plane. vManage pushes path policy; IoT Ops Dashboard monitors the IR fleet; SEA gives engineers zero-trust access to reconfigure a pod router remotely; Meraki Dashboard manages sensors/cameras; RESTCONF (proven in `catalyst_restconf.py`) provisions VLANs/QoS zero-touch — the citizens' segment is stamped by code, not console cables.

5. Pitch Framing — Three Tiers, One Sentence Each

Tier 1 — The Edge: IW9167E gives every victim a Wi-Fi bubble that needs nothing from the city. **Tier 2 — The Mesh:** URWB and LoRa guarantee that pods survive *together* — kilometers of invisible fiber and a sub-GHz whisper that penetrates rubble. **Tier 3 — The Brain:** IR1800 + IOx runs our actual Node.js triage at the edge, SD-WAN guarantees the path, ThousandEyes sees the failure before it happens.

And the closing line: *"Everything you just watched in Docker is this rack of hardware. The containers were never the product — they were the rehearsal."*

6. Rejected — and Why (say this out loud; it builds credibility)

Product	Why not
Nexus 9000/8000, Silicon One G300	Hyperscale data-center silicon. Our EOC is a tent with two UCS boxes, not a leaf-spine fabric.
Catalyst 9500/9600 core, 9800 WLC	Campus core for thousands of users; a pod has ~6 devices and one AP.
HyperFlex / Intersight	Hyperconverged infra assumes a DC ops team; a disaster has none.
Crosswork	Telco-scale automation; wrong operator, wrong scale.
AppDynamics / Splunk APM	ThousandEyes + our own dashboards cover observability at POC scale.
ISE (802.1X identity)	Deliberately rejected at the citizen edge: victims must connect with zero credentials . Open captive portal is a feature. (Keep identity for the <i>management</i> plane via SEA.)
Umbrella full SASE stack	DNS-layer filtering on the portal: yes. Full SSE/CASB: no cloud apps to broker in a disaster zone.

7. Honesty Ledger (if a judge asks "what's real?")

- **Real today:** the entire application layer (14 services), real Meraki Dashboard API telemetry (`meraki_live.py`), real Catalyst 8000v RESTCONF configuration (`catalyst_restconf.py`), real Webex bot notifications, real Ed25519 offline verification, real failover behavior against simulated links.
- **Simulated today:** the links themselves (link-node containers with latency + loss physics), the MT sensor fleet (sensor-simulator, same data contract as the real webhook), infrastructure failure (simulation-controller stopping containers).
- **Hardware-only (this document):** URWB radios, LoRa thin pipe, SRST voice, MV72X snapshots, SD-WAN/ThousandEyes replacing the simulated link layer. Each has a named code seam where it plugs in.